be an end-to-end security protocol for API requests in IoT environments. OSCORE is based on the use of pre-shared keys between the client and server and makes use of CoAP and COSE so that cryptographic algorithms and message formats are suitable for constrained devices.

> **NOTE** OSCORE can be used either as an alternative to transport layer security protocols such as DTLS or in addition to them. The two approaches are complimentary, and the best security comes from combining both. OSCORE doesn't encrypt all parts of the messages being exchanged so TLS or DTLS provides additional protection, while OSCORE ensures end-to-end security.

To use OSCORE, the client and server must maintain a collection of state, known as the security context, for the duration of their interactions with each other. The security context consists of three parts, shown in figure 13.4:

- A *Common Context*, which describes the cryptographic algorithms to be used and contains a Master Secret (the PSK) and an optional Master Salt. These are used to derive keys and nonces used to encrypt and authenticate messages, described later in this section.
- A *Sender Context*, which contains a Sender ID, a Sender Key used to encrypt messages sent by this device, and a Sender Sequence Number. The sequence number is a nonce that starts at zero and is incremented every time the device sends a message.